

Pattern detection and shape analysis for time series

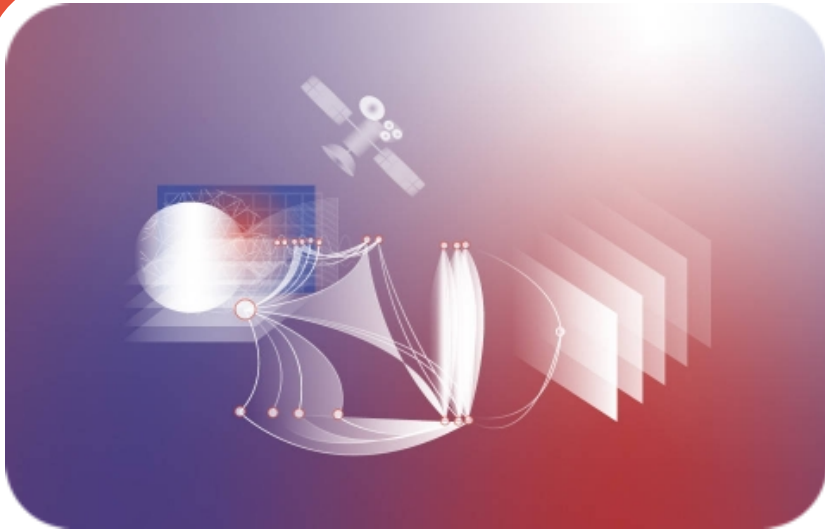
Thibaut Germain

Supervisors:

Charles Truong & Laurent Oudre

June 23rd, 2025

- Centre Borelli is a **multidisciplinary** research unit that brings together researchers in mathematics, computer science and neuroscience.



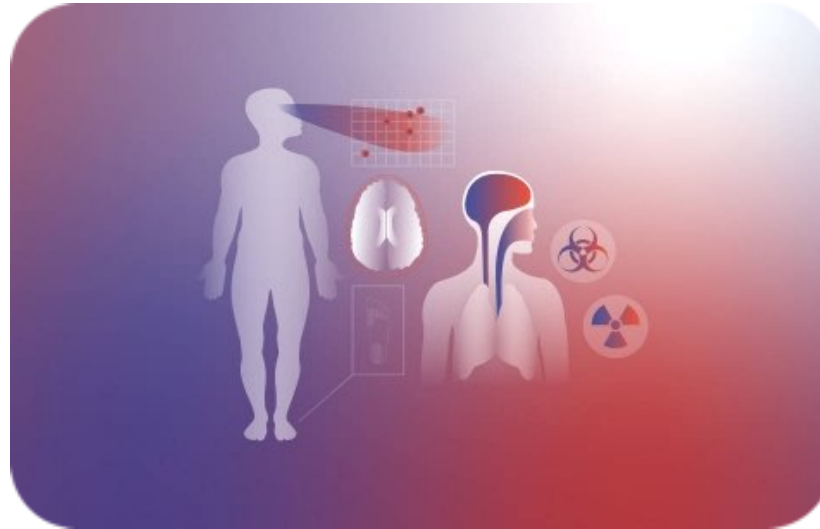
Mathematical foundations and algorithms for artificial perception and learning



L. Oudre



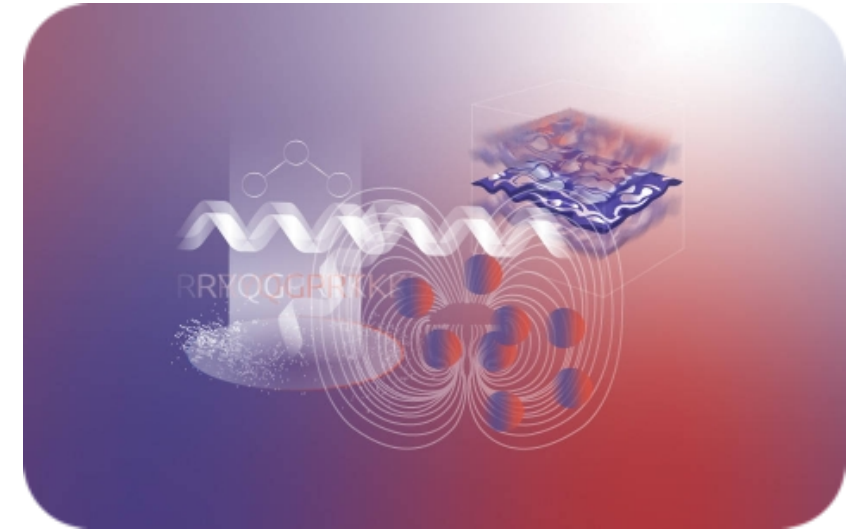
C. Truong



Integrative and behavioral neurosciences in humans and animals

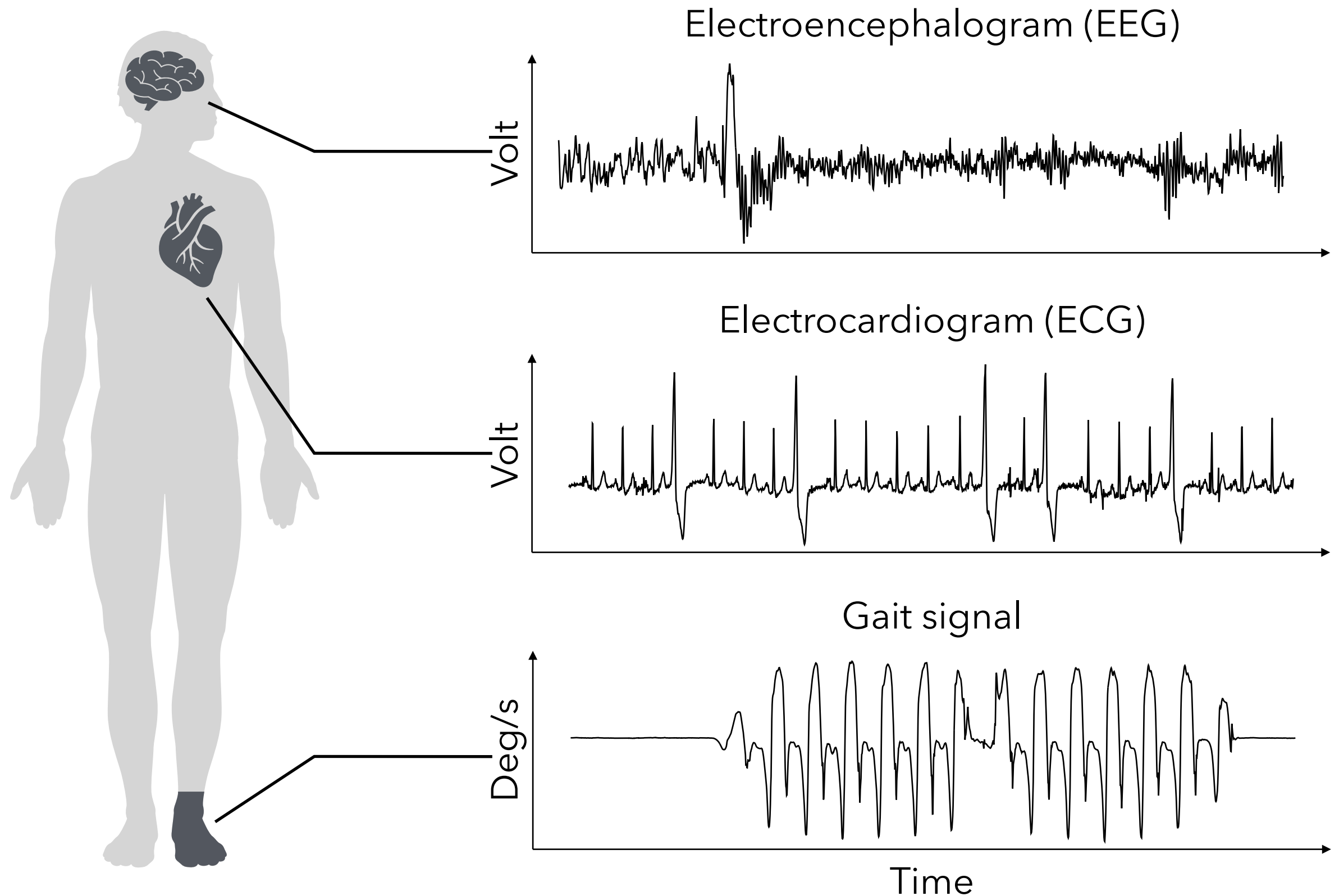


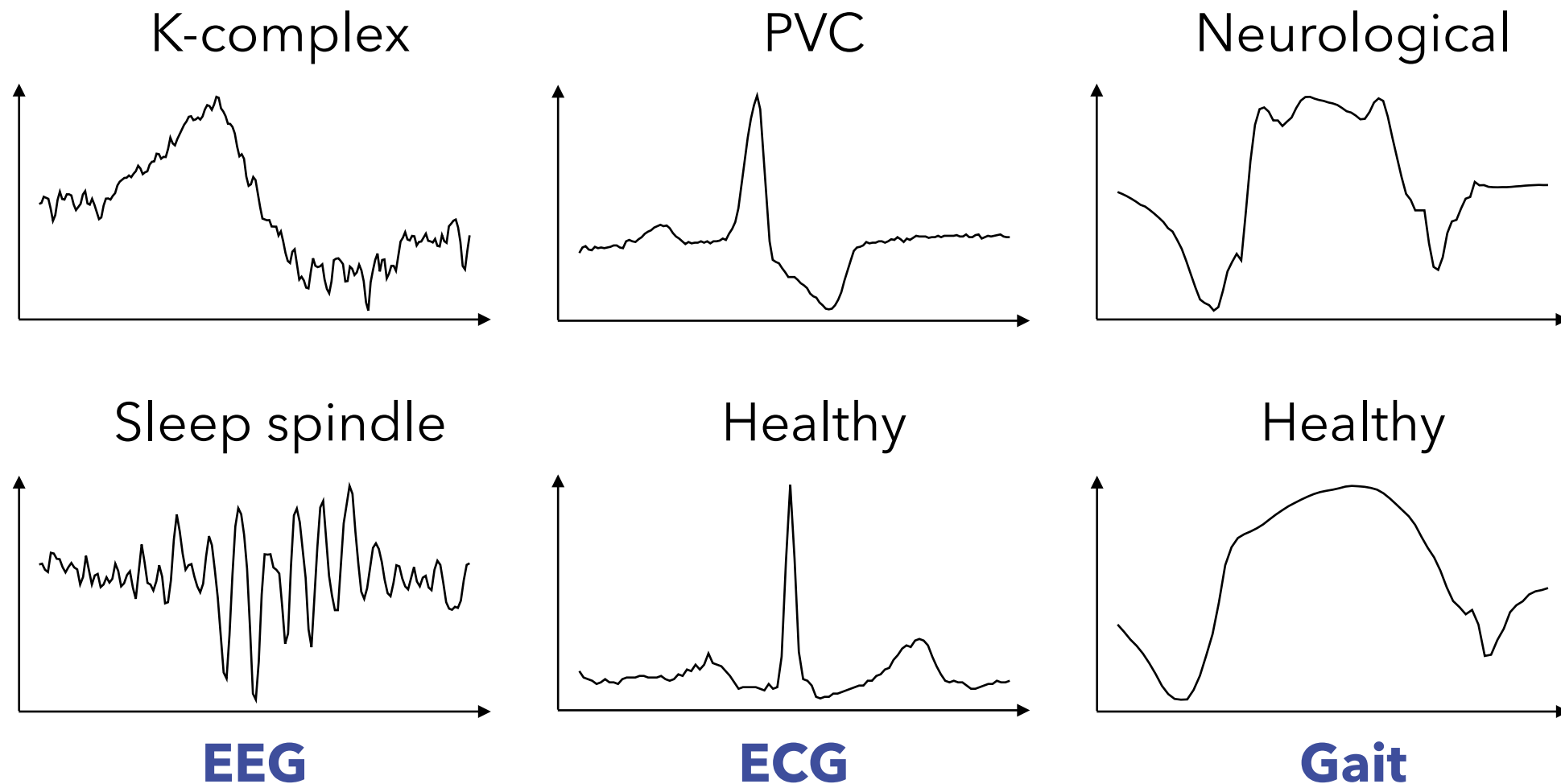
E. Krejci



Modeling, analysis and simulation of physical and biological phenomena

Mouse ventilation analysis





- ▶ Biomedical time series often exhibit patterns that:
 - Consistently observed across different subjects.
 - Reflect the physiological states of subjects.
- ▶ Biomedical patterns are appealing features for statistical analysis (diagnosis, longitudinal follow-up, etc).

Historical approach:

- ▶ Extracting and comparing key features from biomedical patterns.

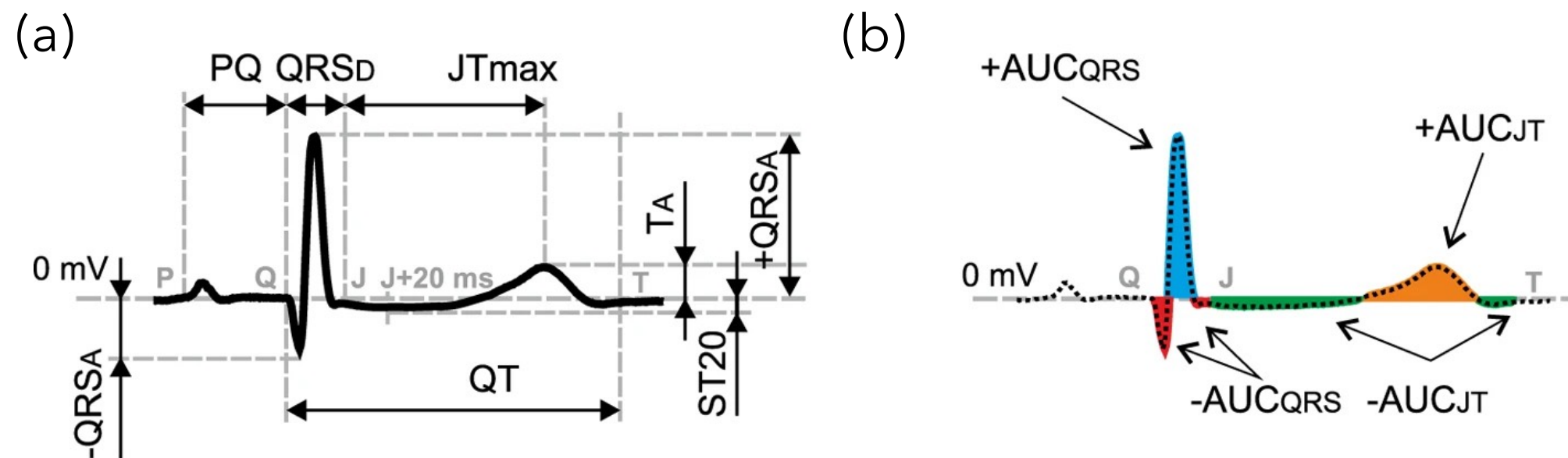
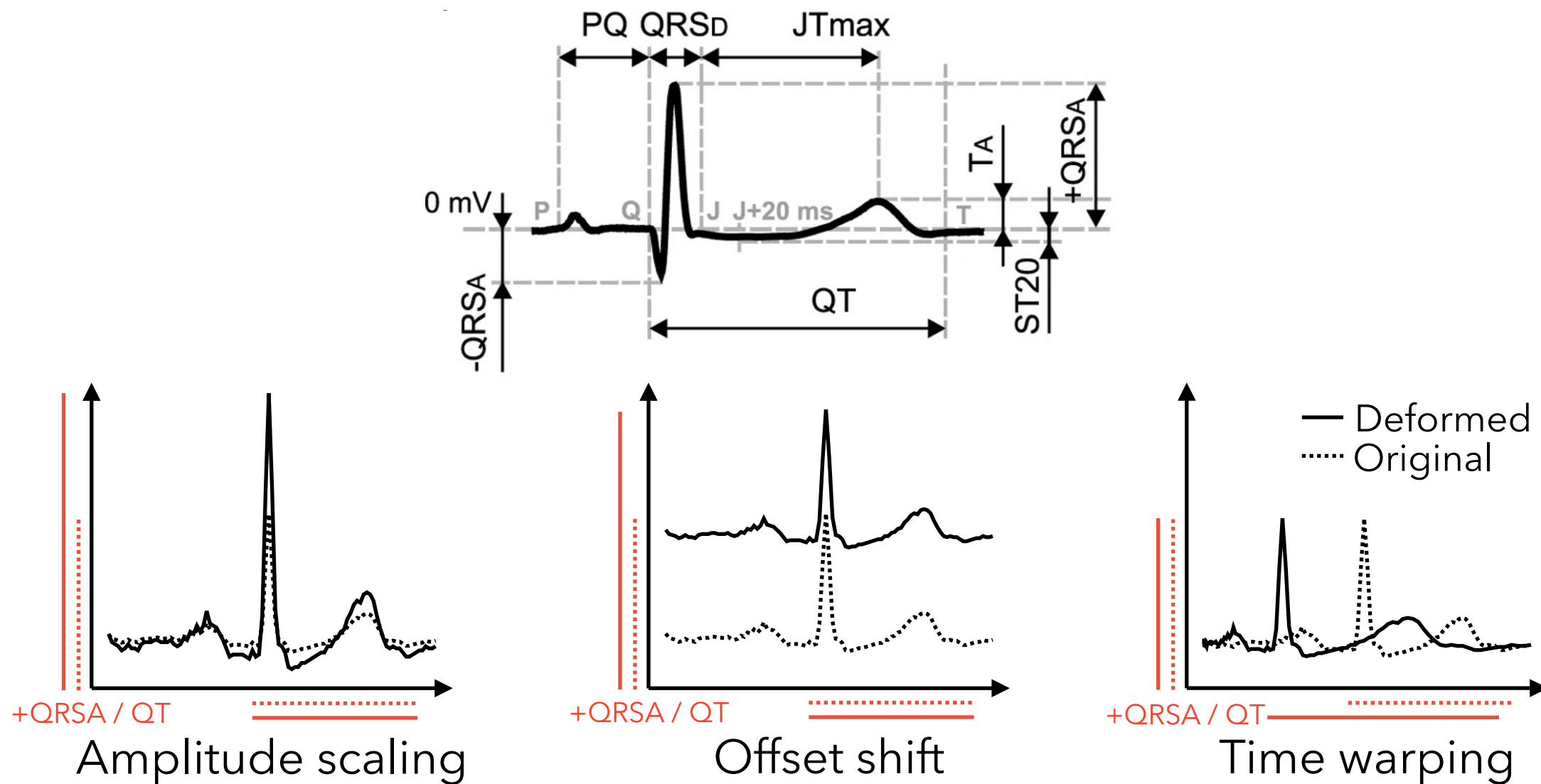


Figure: From [Mar+17], ECG features accounting for (a) intervals' duration and peaks' height, (b) area under the curve of specific intervals.

Issues:

- ▶ Features partially describe the shape of a pattern.
- ▶ Features sensitive to pattern deformations.



Desirable analysis:

- ▶ Comparing the shape of objects independently of some deformations.
- ▶ Comparing the shape of objects by quantifying relevant deformations.

Such problematics are specifically addressed by the research community

Shape analysis

- Shape analysis finds its roots in biology [Tho17, Gre98],

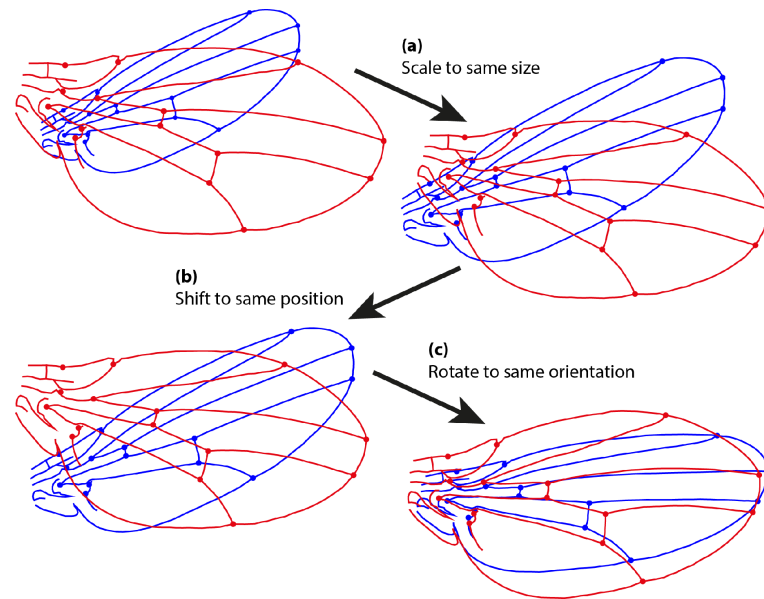


Figure: From [Kli15].

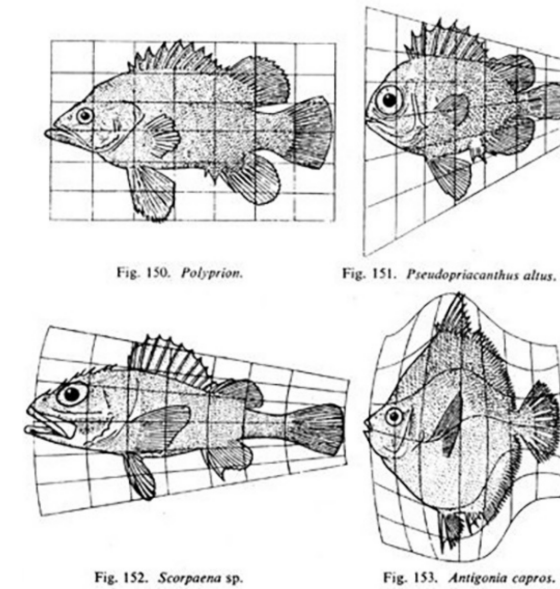


Figure: From [Tho17]

- A strong theoretical background [Boo86, Ken77, Gre94, You10, Bau21],
- Finds applications in medical analysis, Alzheimer [Deb19], glaucoma [Lee17].

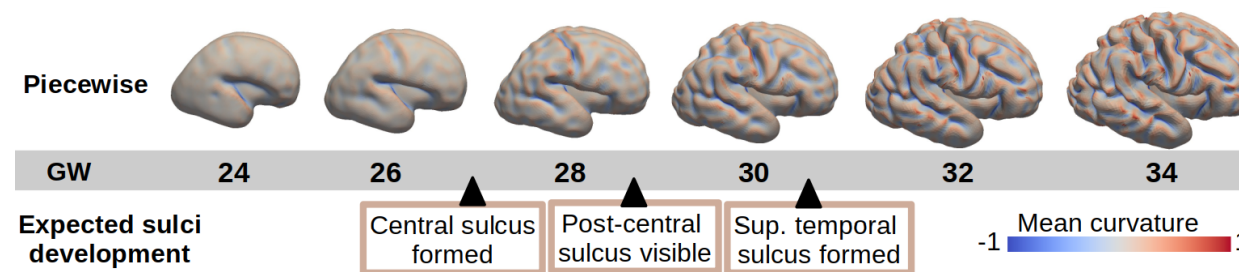


Figure: From [Gau23], analysis of fetal brain development.

- Shape analysis is primarily concerned with images and must be refined to encompass temporal dynamics as needed in time series.

A **Group action** models:

- ▶ The deformation by a group **G**.
- ▶ The geometric objects (time series) by a set **M**.
- ▶ The action of $g \in G$ on $m \in M$ by the deformed object $g \cdot m \in M$.

Two **Time series** representations:

- ▶ Functional: It is a function f from a closed interval $I \subset \mathbb{R}$ taking value in $\mathbb{R}^d$.
- ▶ Discrete: It is a sequence $(f(t_1), \dots, f(t_n)) \in \mathbb{R}^{n \times d}$ sampled a time $t_1 < \dots < t_n \in I$.

Example:

The group of linear trends $G = \{t \in [0,1] \mapsto \alpha t + \beta \in \mathbb{R} \mid (\alpha, \beta) \in \mathbb{R}^2\}$ acts on the set of time series $M = L^2([0,1], \mathbb{R})$ by the action:

$$\begin{array}{ccccc} G & & \times & & M \\ \downarrow & & \bullet & & \downarrow \\ \text{graph of } g & & & & \text{graph of } m \\ & & & & \text{graph of } g \cdot m = m + g \end{array}$$

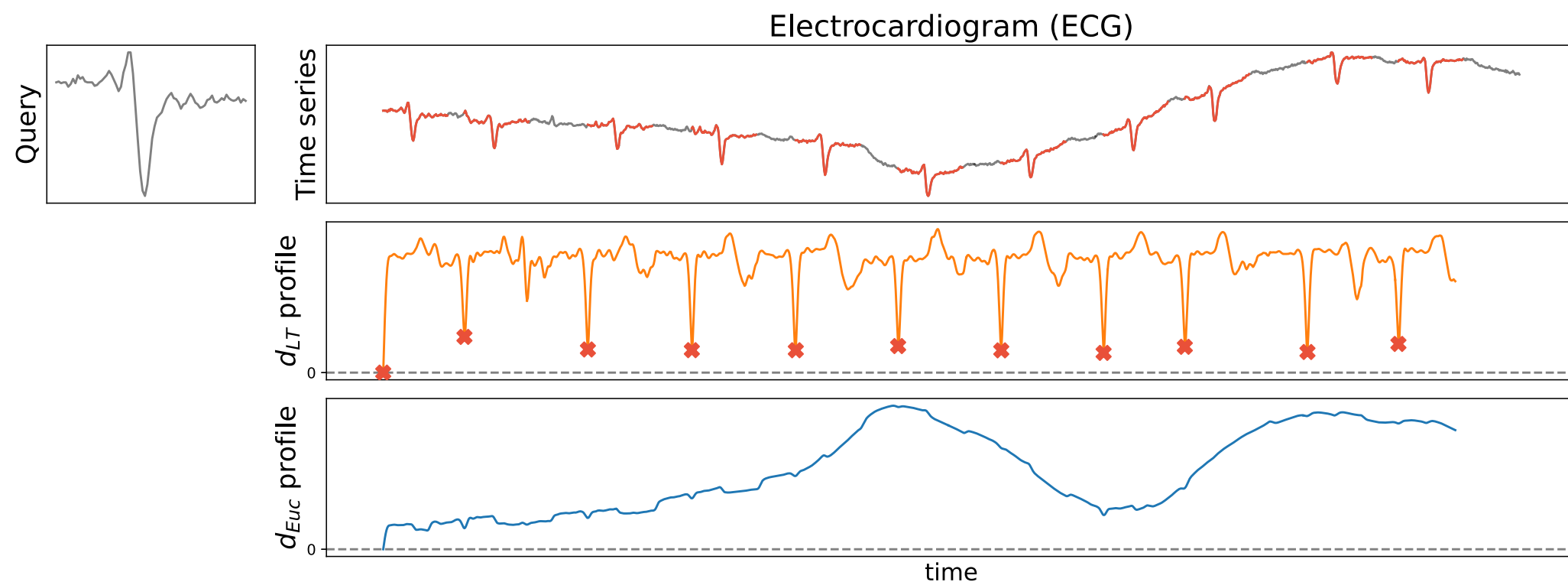
The diagram illustrates the group action of linear trends on time series. It shows a red line g (representing a linear trend) being added to a blue curve m (representing a time series) to produce a new blue curve $g \cdot m = m + g$. The action is represented by the symbol $\times$ (or $\bullet$) and the mapping $\mapsto$ (or $=$).

Aim: Retrieving all occurrences of a query sequence in a time series.

Shape requirement: The similarity functions might need to be invariant to some deformations.

Contribution:

1. Metric invariant to linear deformations [Ger24a].



2. Convolution locally invariant to linear deformations [Ger25].

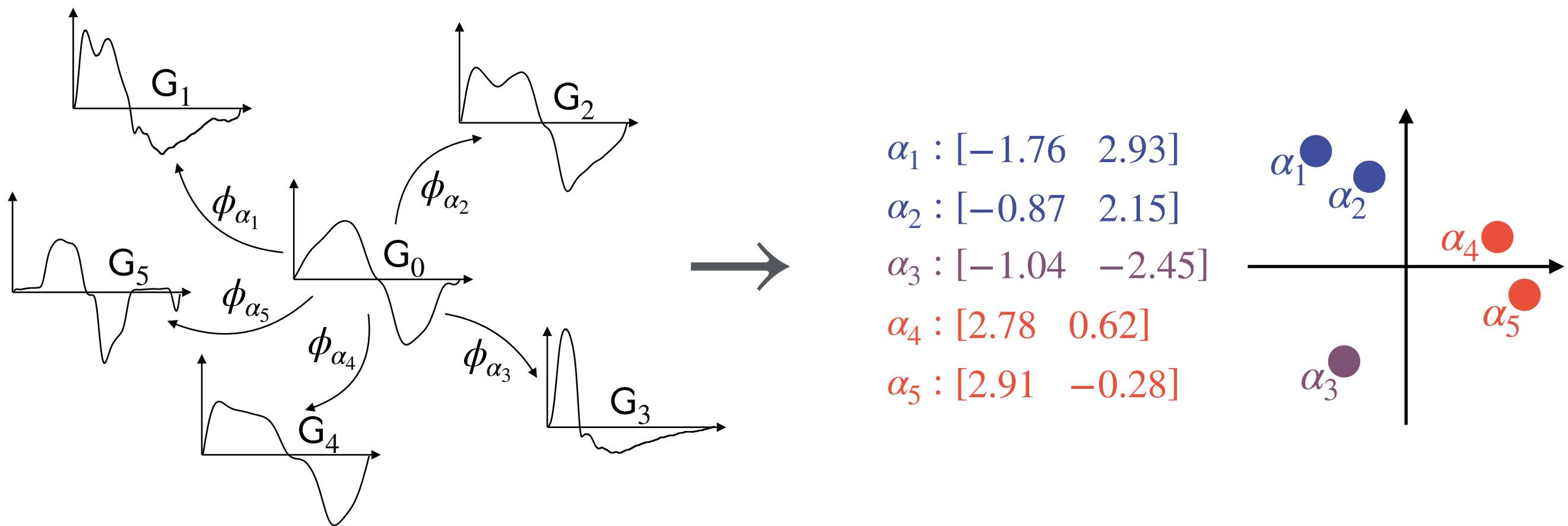
[Ger24a] [Thibaut Germain](#), Charles Truong, and Laurent Oudre. “**Linear-trend normalization for multivariate subsequence similarity search**”. In: 2024 IEEE 40th International Conference on Data Engineering Workshops (**ICDEW**). IEEE, 2024, pp. 167–175. DOI: [10.1109/ICDEW61823.2024.00028](https://doi.org/10.1109/ICDEW61823.2024.00028)

[Ger25] [Thibaut Germain](#), Chrysoula Kosma, and Laurent Oudre. “**Time series representations with hard-coded invariances**”. In the 42nd International Conference on Machine Learning (**ICML**).

Aim: Embed time series irregularly sampled and of variable length as vectors accounting for time series shape structure.

Shape requirement: Transitive group action with a relevant group of deformations for time series.

Contribution: An unsupervised framework for learning time series embedding suited for shape-related statistical analysis [Ger24d].



Local invariance to linear deformation



Aim:

Propose metrics and **convolutions** (locally) **invariant** to linear deformations computationally **scalable**.

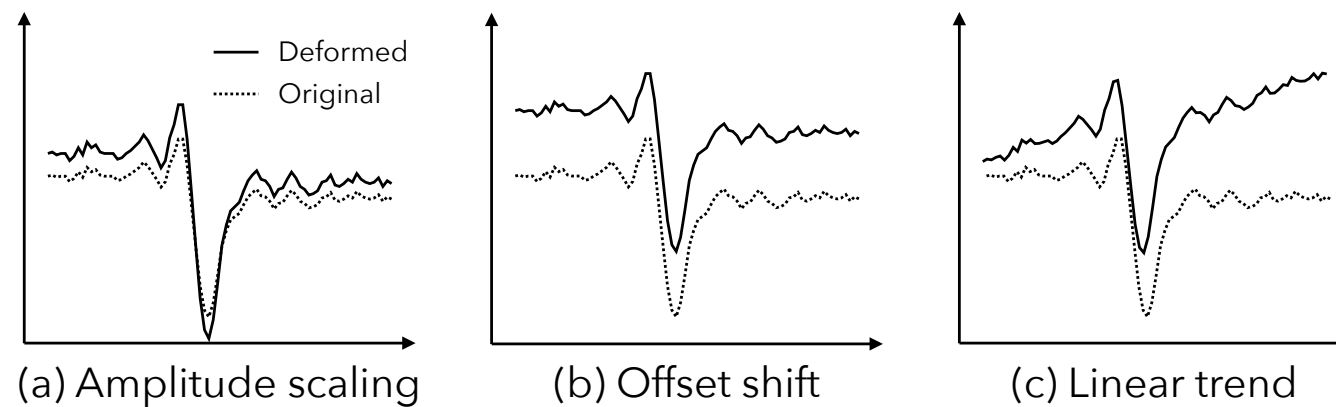
Methodology:

Construct a time series **embedding invariant** to linear deformations and scalable thanks to the **Fast Fourier Transform**.

Application:

Similarity search, Classification, Domain adaptation

Question: How to model local linear deformations and their action ?



Group Action: Consider the group action:

$$(\mathbb{R}_+^* \ltimes H) \times L^2(I, \mathbb{R}^D, \mu) \mapsto L^2(I, \mathbb{R}^D, \mu)$$

$$((\lambda, h), f) \mapsto \lambda f + h$$

with:

$$L^2(I, \mathbb{R}^D, \mu) = \{f : I \subset \mathbb{R} \rightarrow \mathbb{R}^D \mid \int_I \langle f(t), f(t) \rangle^2 \mu(dt) < \infty\} ,$$

and:

H a finite dimensional subspace of $L^2(I, \mathbb{R}^D, \mu)$.

Orbit: Consider $f \in L^2(I, \mathbb{R}^D, \mu)$, its orbit is the set $[f] = \{\lambda f + h \mid (\lambda, h) \in \mathbb{R}_+^* \ltimes H\}$.

Next question: How to define a deformation-invariant metric ? That is a metric on the set of orbits, i.e. :

$$d([f], [g]) = ?$$

Aim: Proposing a embedding $L : \mathbf{M} \mapsto \mathbf{M}$ that is:

1. $(\mathbb{R}_+^* \ltimes H)$ -invariant: $L(\lambda f + h) = L(f)$
2. orbit-injective: $L(f) \neq L(g)$ if $[f] \neq [g]$

Proposition 1: The embedding $L = I_d - P_H$ is H -invariant and orbit-injective.
** orthogonal projector*

$$* P_H : m \in \mathbf{M} \mapsto \sum_{i=1}^N \langle m, h_i \rangle_L h_i \in H \text{ with the orthonormal basis } (h_i)_{i \in \llbracket 1, N \rrbracket}$$

Proposition 2: The embedding

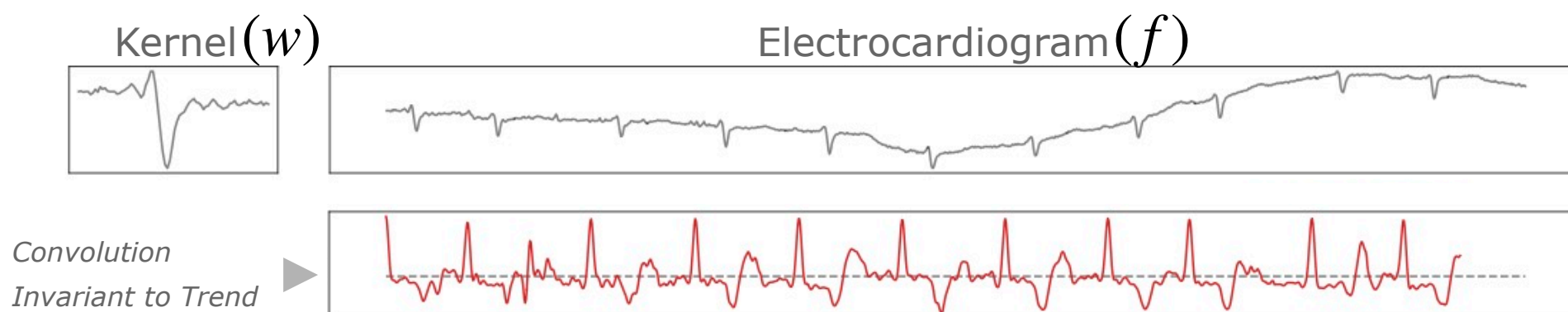
$$\hat{L} : m \in \mathbf{M} \mapsto \begin{cases} L(m) / \|L(m)\|_{\mathbf{M}} & \text{if } m \in \mathbf{M} \setminus H \\ 0_{\mathbf{M}} & \text{else} \end{cases}$$

is $(\mathbb{R}_+^* \ltimes H)$ -invariant and orbit-injective.

Proposition 3 (Invariant metric): The application:

$$d([f], [g]) = \|\hat{L}(f) - \hat{L}(g)\|_M = \sqrt{2 \left(1 - \langle \hat{L}(f), \hat{L}(g) \rangle \right)},$$

defines a metric on the set of orbits (quotient space).



Proposition 4 (Locally invariant convolution): The application:

$$f \circledast^G w : u \in \mathbb{R} \mapsto \int_I \langle (K_u^G f)(t), w(t) \rangle \mu(dt) \in \mathbb{R},$$

where $K_u^G f = \hat{L}(t \in I \mapsto f(t + u))$ is locally invariant to deformations.

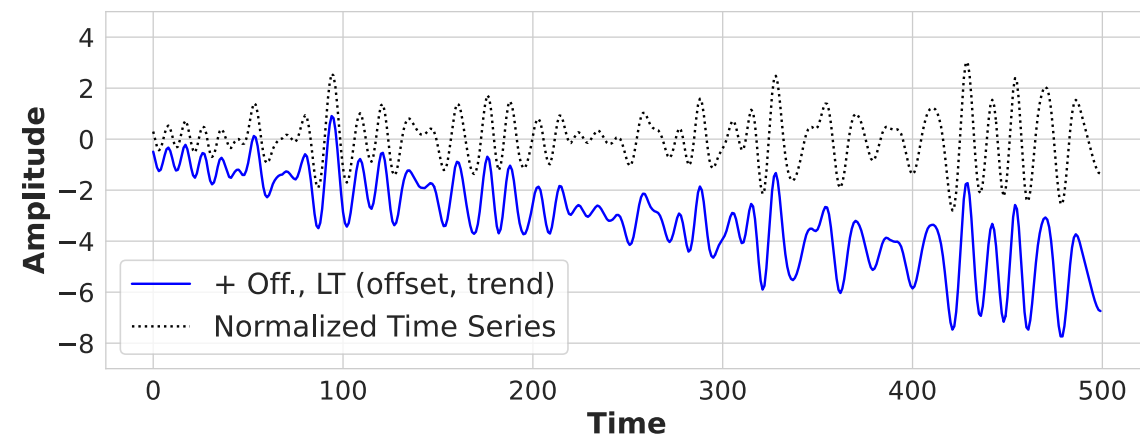
Computational complexity: Given an orthonormal basis of :

$$\langle \hat{L}(f), w \rangle = (\langle f, w \rangle - \sum_{i=1}^N \langle f, h_i \rangle \langle w, h_i \rangle) / \|L(f)\| \implies \mathcal{O}(NDL \log(L)) \text{ (FFT [Mat13])}$$

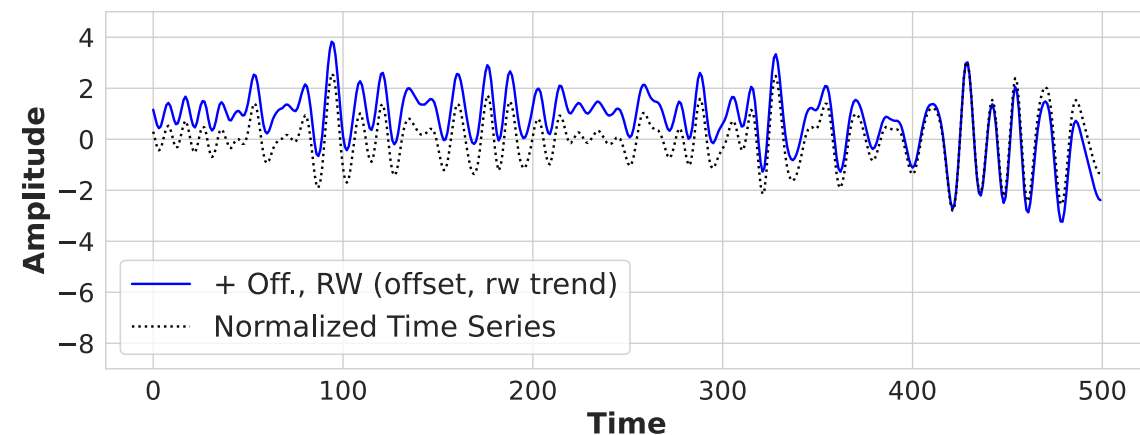
Classification Accuracy (%)

		CONV - normal -	INV. CONV - offset -	INV. CONV - trend -	TS-TCC (FT) (2021) - offset, trend -
HandOutlines	<u>Normalized</u>	77.84 ± 0.0	72.43 ± 0.54	70.54 ± 0.71	88.96 ± 1.17
	+ off.	41.08 ± 0.47	72.43 ± 0.54	70.72 ± 1.02	47.83 ± 0.54
	+ LT	37.39 ± 0.16	71.08 ± 1.08	70.90 ± 0.82	40.87 ± 1.41
	+ off., LT	35.68 ± 0.47	70.45 ± 1.22	70.63 ± 0.41	37.07 ± 0.87
	+ off., RW	35.95 ± 0.0	62.34 ± 1.02	64.59 ± 1.24	35.82 ± 0.15
MixedShapesRegularTrain	<u>Normalized</u>	93.91 ± 0.39	88.48 ± 1.79	92.26 ± 0.70	93.44 ± 1.51
	+ off.	38.96 ± 0.99	88.41 ± 1.83	92.24 ± 0.73	92.56 ± 1.31
	+ LT	29.33 ± 2.16	84.63 ± 3.92	91.55 ± 0.69	82.96 ± 4.72
	+ off., LT	28.22 ± 1.49	84.05 ± 4.08	91.64 ± 0.80	81.35 ± 4.08
	+ off., RW	22.94 ± 0.71	76.98 ± 4.62	90.59 ± 0.64	68.56 ± 4.86
NonInvasiveFetalECGThorax1	<u>Normalized</u>	91.35 ± 0.05	86.53 ± 0.06	85.00 ± 0.33	84.19 ± 1.18
	+ off.	15.32 ± 0.44	85.11 ± 0.73	83.84 ± 0.12	71.47 ± 6.51
	+ LT	8.09 ± 0.23	42.49 ± 0.33	82.56 ± 0.45	49.54 ± 5.97
	+ off., LT	6.09 ± 0.17	42.36 ± 0.65	82.02 ± 0.52	47.61 ± 6.70
	+ off., RW	3.48 ± 0.14	37.22 ± 0.94	73.33 ± 0.05	31.55 ± 6.16
FordB	<u>Normalized</u>	82.35 ± 0.57	83.33 ± 0.5	82.84 ± 0.65	78.37 ± 0.30
	+ off.	60.91 ± 3.17	81.61 ± 1.38	83.37 ± 0.31	78.29 ± 0.43
	+ LT	53.83 ± 1.10	70.50 ± 6.02	82.68 ± 0.91	76.85 ± 2.23
	+ off., LT	53.04 ± 0.75	69.47 ± 7.34	82.35 ± 0.66	76.49 ± 2.83
	+ off., RW	50.91 ± 0.26	68.80 ± 8.21	80.37 ± 0.43	75.77 ± 1.67
Yoga	<u>Normalized</u>	76.12 ± 0.24	76.80 ± 1.69	76.92 ± 0.14	80.64 ± 0.63
	+ off.	53.73 ± 0.26	72.07 ± 1.93	75.71 ± 0.18	74.97 ± 1.59
	+ LT	52.86 ± 0.14	67.71 ± 2.99	74.17 ± 0.29	62.21 ± 0.74
	+ off., LT	51.13 ± 0.07	64.85 ± 2.05	72.88 ± 0.38	61.85 ± 1.26
	+ off., RW	48.82 ± 0.12	65.67 ± 0.74	72.58 ± 0.19	62.69 ± 2.64
Percentage Drop (%) with respect to <u>Normalized</u> (lower the better in abs. value)	+ off.	-48% ± 20%	-1% ± 2%	0% ± 0%	-13% ± 17%
	+ LT	-55% ± 22%	-16% ± 17%	-1% ± 1%	-26% ± 19%
	+ off., LT	-57% ± 22%	-18% ± 17%	-1% ± 2%	-28% ± 20%
	+ off., RW	-59% ± 23%	-23% ± 16%	-6% ± 4%	-34% ± 22%

Figure. FordB dataset (UCR archive), with the normalized input (dotted line) and deformed versions of the series (in blue).



- Addition of random offset and linear trend -

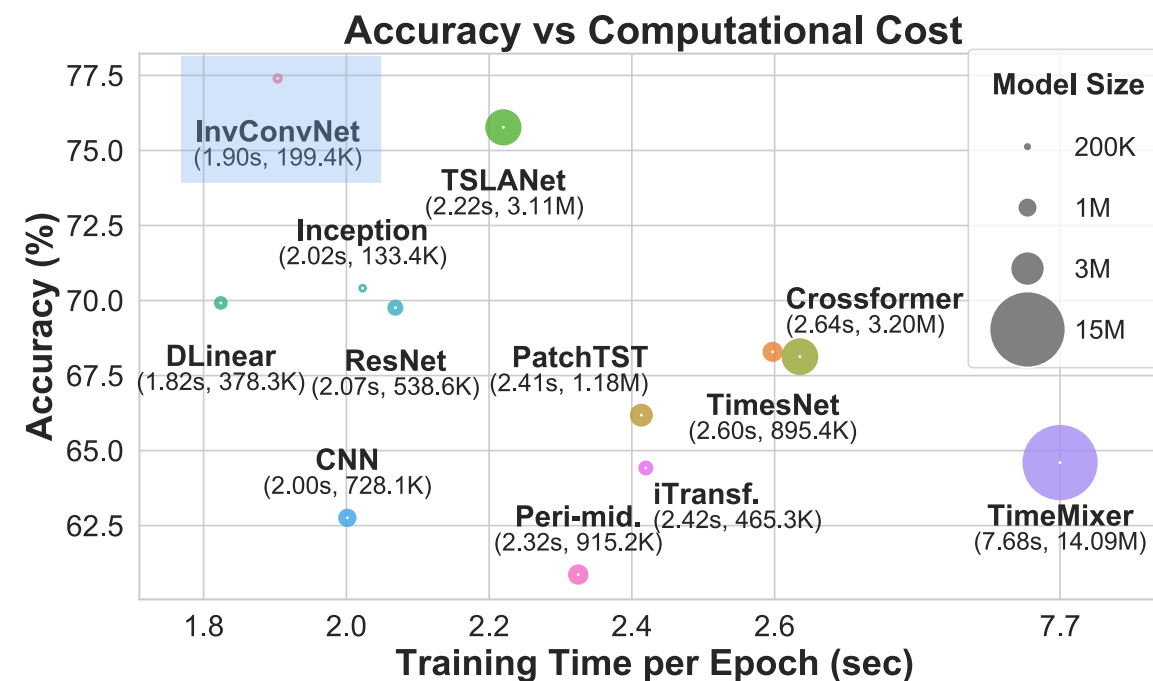
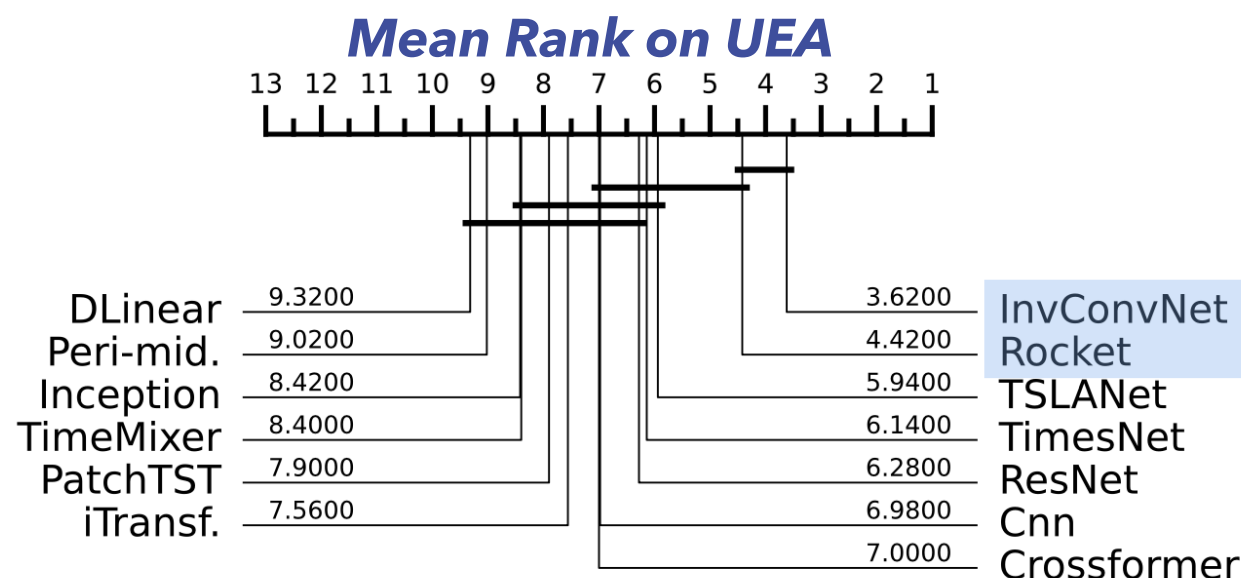


- Addition of random offset and random walk trend -

Settings:

- **Task:** Classification on five UCR datasets [Dau19], trained on raw data and tested on 4 unseen synthetic deformations.
- **Models:** Self-supervised TS-TCC (pre-trained on deformations) vs CNNs with different filter types: normal (non-invariant), offset-invariant, and trend-invariant.

Classification Benchmark:



Classification Accuracy (%)

Datasets	INVCONVNET (ours)	TIMESNET (2022)	PATCHTST (2022)	CROSSFORMER (2022)	ITRANSF. (2023)	PERI-MID. (2024)	TSLANET (2024)	DLINear (2023)	TIME MIXER (2024a)	INCEPTION (2020)	RESNET (2017)	CNN (2018)	ROCKET (2020)
UEA (26 datasets)	71.81 ± 0.80	66.87 ± 1.72	66.18 ± 1.26	66.37 ± 1.35	64.42 ± 2.05	60.87 ± 3.00	68.70 ± 1.19	61.51 ± 1.05	64.60 ± 1.69	62.86 ± 1.96	67.37 ± 1.59	65.67 ± 1.64	71.29 ± 0.90
UCIHAR	96.63 ± 0.49	91.66 ± 0.62	85.74 ± 0.50	93.43 ± 0.56	94.13 ± 0.03	92.64 ± 0.57	94.71 ± 0.66	57.47 ± 0.73	84.40 ± 1.07	95.26 ± 0.55	96.04 ± 0.48	95.78 ± 0.20	92.06 ± 0.15
Sleep-EDF	84.95 ± 0.39	74.64 ± 0.73	78.53 ± 0.28	79.82 ± 0.89	53.11 ± 0.21	63.50 ± 1.52	84.98 ± 0.43	36.15 ± 0.21	74.07 ± 2.05	84.06 ± 0.39	85.62 ± 0.13	82.41 ± 0.56	83.88 ± 0.09
Epilepsy	98.43 ± 0.04	97.62 ± 0.20	98.01 ± 0.05	98.23 ± 0.12	97.80 ± 0.26	97.99 ± 0.29	98.23 ± 0.05	82.26 ± 0.06	97.94 ± 0.18	97.65 ± 0.20	98.16 ± 0.04	97.61 ± 0.28	98.38 ± 0.02

Transfert Learning:

Classification Accuracy (%)

Methods	Train → Test												Avg. Acc. (%)
	A → B	A → C	A → D	B → A	B → C	B → D	C → A	C → B	C → D	D → A	D → B	D → C	
TS-TCC (FT)	55.33 ± 1.44	52.52 ± 4.55	62.13 ± 1.39	48.05 ± 3.32	71.50 ± 1.83	100.0 ± 0.0	40.76 ± 2.22	98.25 ± 1.22	99.34 ± 0.50	46.98 ± 0.65	100.0 ± 0.0	74.28 ± 2.77	70.76 ± 1.66
TS2VEC (FT)	54.11 ± 1.46	54.07 ± 1.91	52.54 ± 1.89	55.06 ± 0.17	88.72 ± 0.47	100.0 ± 0.0	57.81 ± 2.18	78.30 ± 3.80	78.41 ± 4.39	60.37 ± 1.95	99.97 ± 0.02	86.82 ± 0.54	72.18 ± 1.57
INVCONVNET (Sup.)	55.90 ± 0.42	55.93 ± 0.34	53.41 ± 0.14	85.10 ± 0.63	78.54 ± 0.17	99.05 ± 0.08	70.75 ± 1.32	85.04 ± 0.13	85.12 ± 0.15	70.91 ± 0.73	100.0 ± 0.0	78.49 ± 0.38	76.52 ± 0.37
INVCONVNET-N (Sup.)	60.55 ± 0.88	55.50 ± 1.82	53.50 ± 0.85	60.26 ± 2.01	77.30 ± 0.60	93.50 ± 0.90	64.93 ± 0.48	84.87 ± 0.23	84.46 ± 0.51	59.98 ± 0.98	99.96 ± 0.0	77.14 ± 0.14	72.66 ± 0.78

Task: training and testing on A, B, C, D sub-datasets of Fault-Diagnosis [Les16]. InvConvNet, InvConvNet-N (normal) are supervised on source and TS-TCC, TS2Vec are self-supervised (fine-tuned on source).

Advantages:

- ▶ Generalisation (Robustness to unseen domains and deformations).
- ▶ Scalable computation.
- ▶ Lightweight models (requires smaller training datasets).

Limits:

- ▶ Preselected sets of deformations.
- ▶ Sensitivity to signals with irregularities.

Future work:

- ▶ Activation layer for selecting the appropriate sets deformation.
- ▶ Learning the set of deformations.

Deformation-based embedding for time series



Aim:

Analyze **inter-individual variability** in a dataset of **irregular sampled** and **variable sequence length** time series.

Methodology:

Unsupervised representation learning with Large Deformation Diffeomorphic Metric Mapping (**LDDMM**) to capture **spatiotemporal structure**.

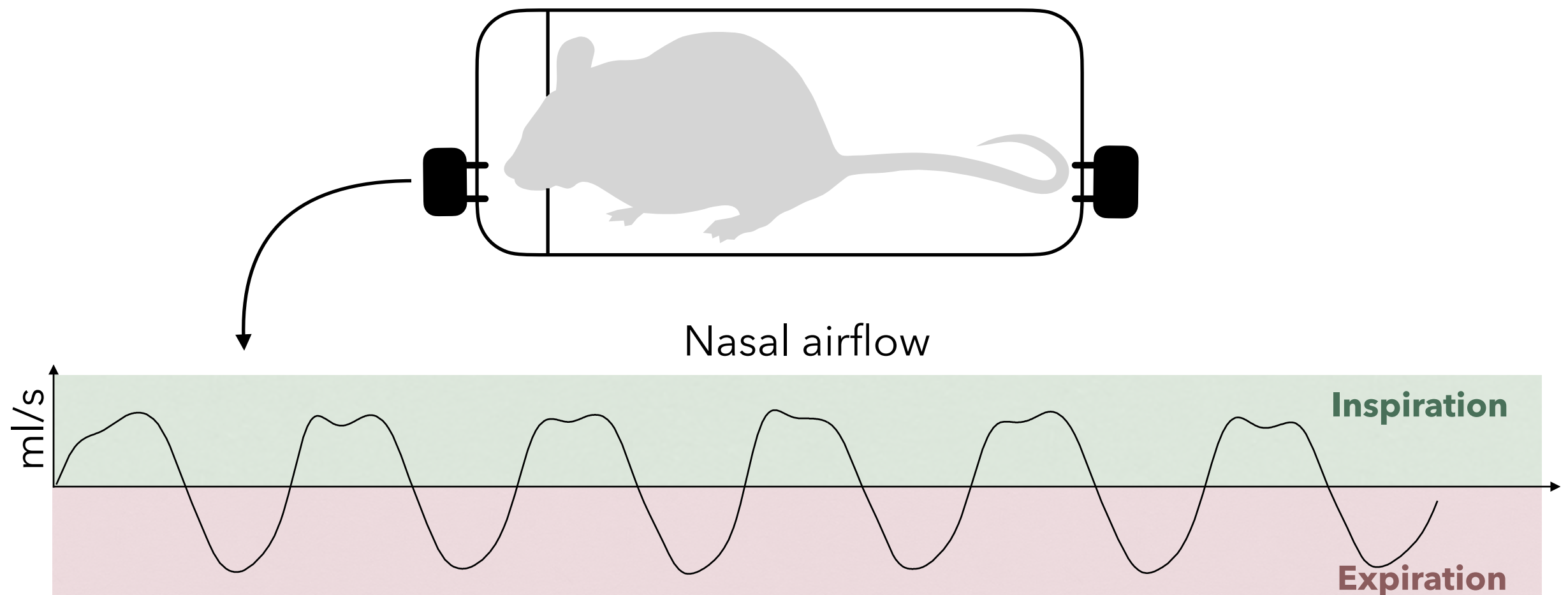
Application:

Mouse ventilation analysis

Aim: Assessing mouse ventilation and its alterations in response to physiological or environmental changes.

Applicability: Studying drugs' effects on the respiratory system, monitoring mouse models of human diseases, evaluating airway irritants, etc.

Measurement: plethysmography

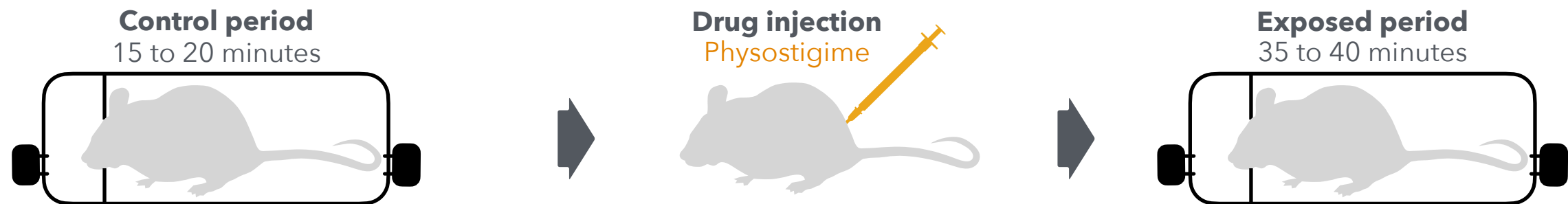


Mouse ventilation analysis

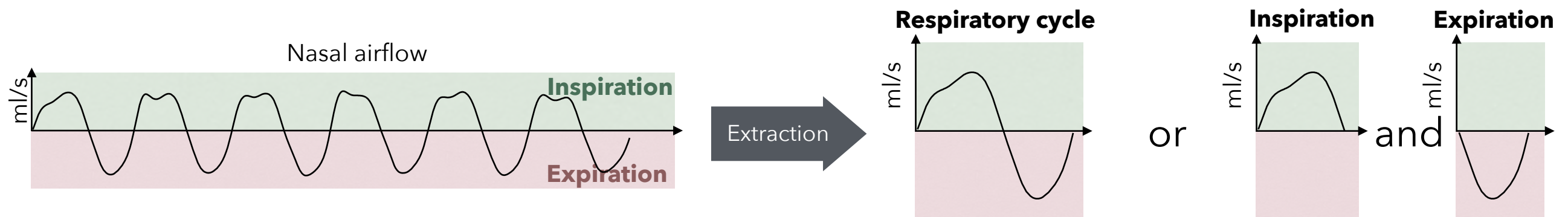


Aim: Assessing ventilation response after the exposure of a neurotoxin for mice of different genotypes.

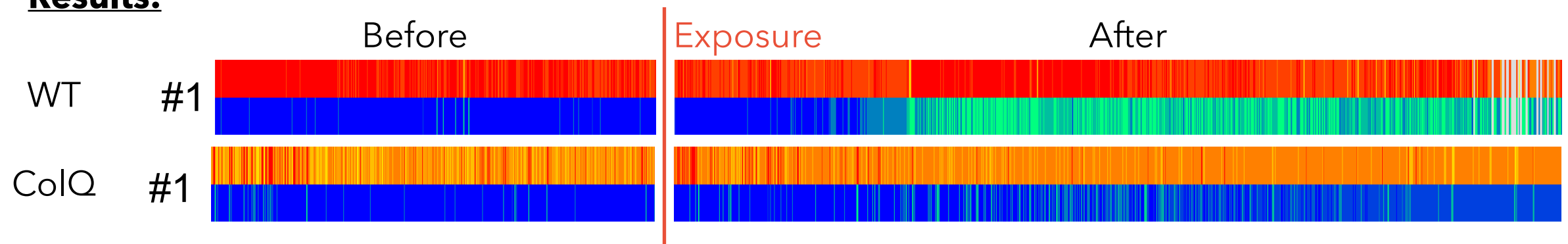
Experimental protocol: For each genotype (WT, ColQ):



Analysis insights: A time series with elementary patterns.



Results:



Input: A time series dataset $\{s_j : \mathbf{I}_j \mapsto \mathbb{R}^d \mid j = 1, \dots, N\}$.

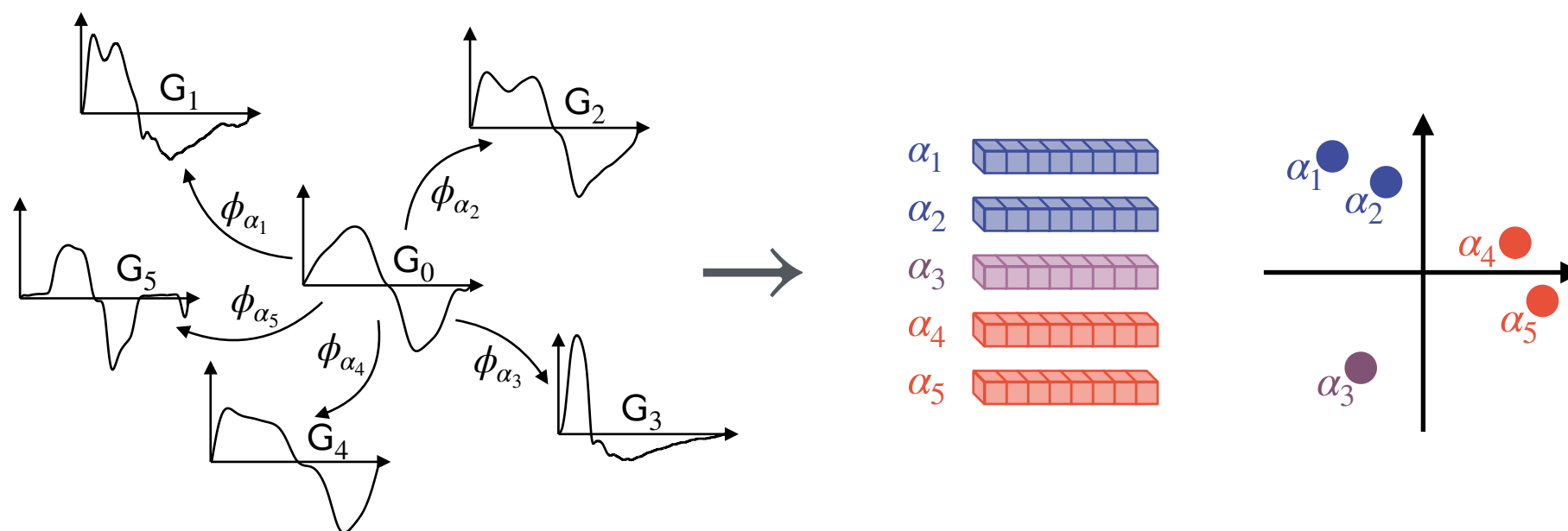
► Any time series s_j is associated to its graph $\mathbf{G}_j = \{(t, s_j(t)) \mid t \in \mathbf{I}_j\}$.

⚠ In practice, it is a sampled graph: $\tilde{\mathbf{G}}_j = \{(t_{i,j}, s_j(t_{i,j})) \mid t_{1,j} < \dots < t_{n_j,j} \in \mathbf{I}_j\}$

► Any $\mathbf{G}_j$ is represented by the parameters $\alpha_j \in \mathbb{R}^m$ of a deformation whose action on a graph $\mathbf{G}_0 = \{(t, s_0(t)) \mid t \in \mathbf{I}\}$ verifies:

$$\mathbf{G}_j \approx \phi_{\alpha_j} \cdot \mathbf{G}_0 = \{\phi_{\alpha_j}(t, s_0(t)) \mid t \in \mathbf{I}\},$$

where $\phi_{\alpha_j} : \mathbb{R}^{d+1} \mapsto \mathbb{R}^{d+1}$ is a diffeomorphism parametrized by $\alpha_j \in \mathbb{R}^m$.



Question: How to learn the reference graph $\mathbf{G}_0$ and the deformations ϕ_{α_j} .

The reference graph $\mathbf{G}_0$ and the deformation ϕ_{α_j} are learned by solving the empirical Fréchet mean problem:

$$\arg \min_{\mathbf{G}_0, (\alpha_j)_{j \in \llbracket 1, N \rrbracket}} \frac{1}{N} \sum_{j=1}^N \left(\underbrace{d_{\mathbf{G}}^2(\phi_{\alpha_j} \cdot \mathbf{G}_0, \mathbf{G}_j)}_{\text{data attachment}} + \underbrace{\lambda d_{\Phi}^2(\text{Id}, \phi_{\alpha_j})}_{\text{regularization}} \right)$$

where $\phi_{\alpha_j} : \mathbb{R}^{d+1} \mapsto \mathbb{R}^{d+1}$ is a diffeomorphism parametrized by $\alpha_j \in \mathbb{R}^m$, and $d_{\mathbf{G}}$ is the varifold loss [Kal17].

Questions:

- ▶ How to parametrize diffeomorphisms?
- ▶ What distance d_{Φ} should be considered?



Theorem [Gla05]: Assuming $v \in L^2([0,1], V)$ a time-varying velocity field in $\mathbb{R}^n$, where V is an RKHS with some regularity assumptions [Gla05]. For any $x_0 \in \mathbb{R}^n$, the differential system:

$$\frac{dX(\tau)}{d\tau} = v_\tau(X(\tau)) \quad \text{with} \quad X(0) = x_0,$$

has a unique solution defined for all $\tau \in [0,1]$.

► At any time $\tau \in [0,1]$, **the flow application:**

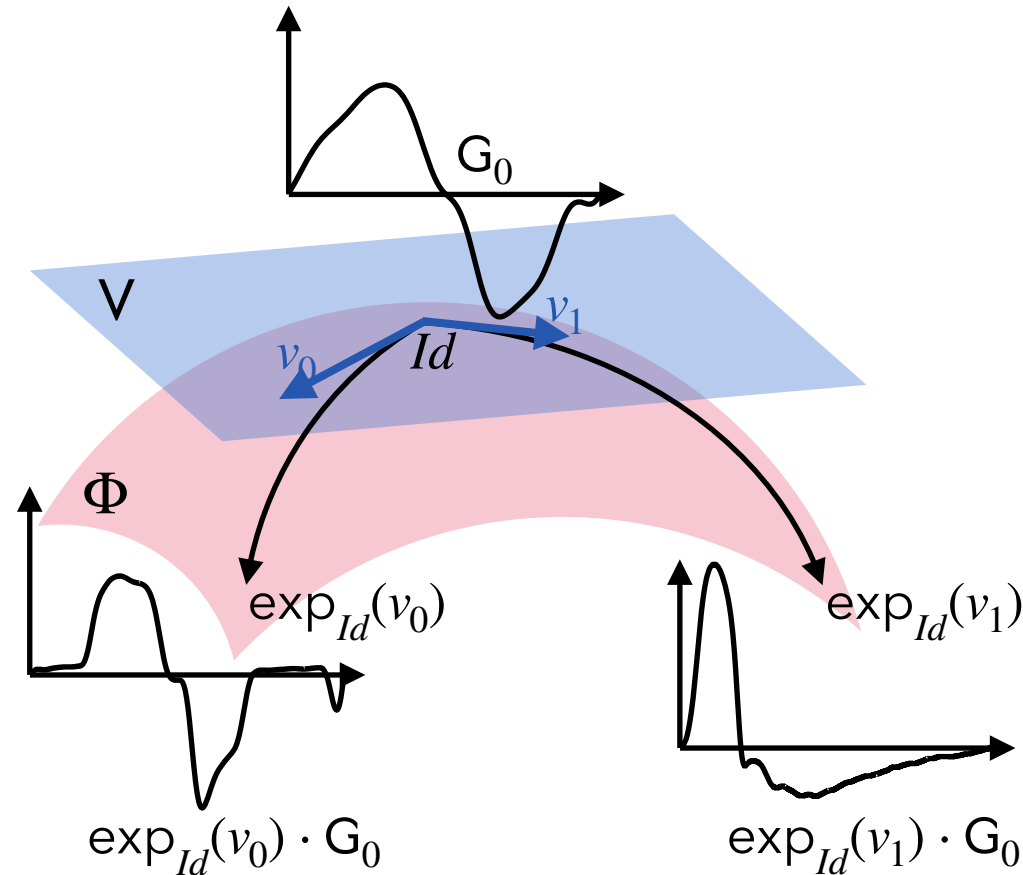
$$\phi_v^\tau : x_0 \in \mathbb{R}^n \mapsto x_0 + \int_0^\tau v_r(\phi_v^r(x_0)) dr \in \mathbb{R}^n$$

is a **diffeomorphism**.

A metric group: The group of diffeomorphisms $\Phi \triangleq \{\phi_v^1 \mid v \in L^2([0,1], V)\}$ is metrizable such that for any $\phi \in \Phi$:

$$d_\Phi^2(Id, \phi) = \inf_{v \in L^2([0,1], V)} \left\{ \int_0^1 \|v_\tau\|_V^2 d\tau \mid \phi_v^1 = \phi \right\},$$

the infimum is reached with a v^* and it conserves its norm along its geodesic path i.e.: $\|v_\tau^*\|_V = \|v_0^*\|_V, \forall \tau \in [0,1]$.



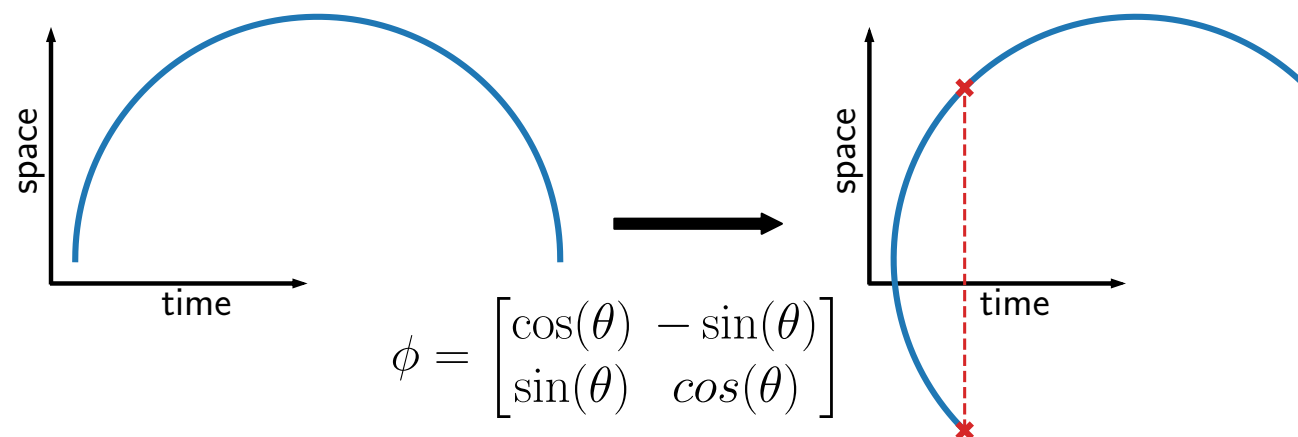
An Exponential map: By denoting $\tau \mapsto \rho_{v_0}(\tau)$ the geodesic starting from Id with initial conditions $v_0 \in V$, the exponential map is [Mil06]:

$$\exp_{Id} : v_0 \in V \mapsto \rho_{v_0}(1) \in \Phi \quad \text{and} \quad d_{\Phi}^2(Id, \exp_{Id}(v_0)) = \|v_0\|_V^2$$

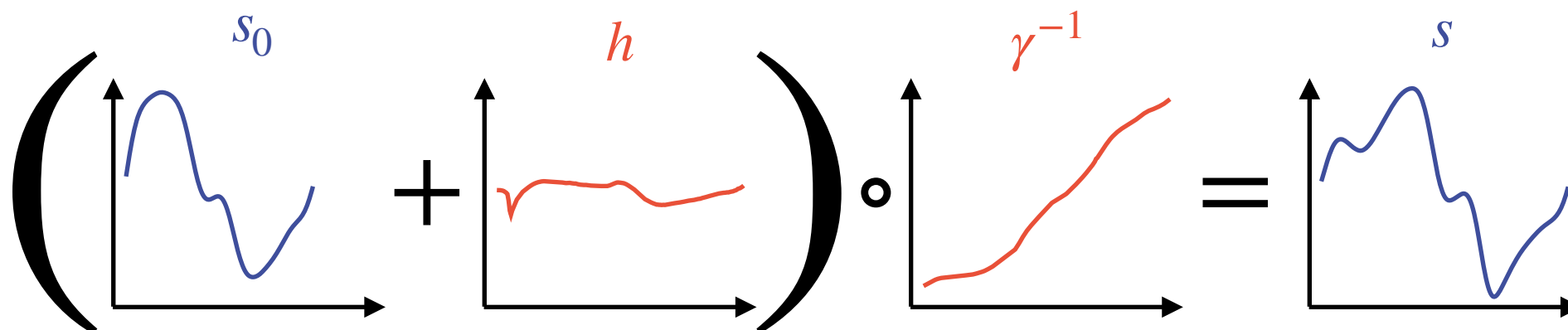
In practice: v_0 is parametrized by K , the kernel associated with V , the sampled time series graph $G_0 \in (\mathbb{R}^{d+1})^{N_0}$ and the parameters $\alpha_0 \in (\mathbb{R}^{d+1})^{N_0}$ such that:

$$v_0 : x \in \mathbb{R}^{d+1} \mapsto \sum_{k \in [N_0]} K(g_0^k, x) \alpha_0^k \in \mathbb{R}^{d+1}$$

Problem: A general diffeomorphism does not necessarily preserve the structure of a time series graph.



Intuition: Let $s_0 : \mathbf{I} \mapsto \mathbb{R}^d$ and $s : \mathbf{J} \mapsto \mathbb{R}^d$, the diffeomorphic deformation ϕ mapping s_0 to s should be seen as distortion $h : \mathbf{I} \mapsto \mathbb{R}$ and a time parametrization $\gamma^{-1} : \mathbf{J} \mapsto \mathbf{I}$ such that: $\phi \cdot G(s_0) = G((s_0 + h) \circ \gamma^{-1}) = G(s)$.



Gaussian Kernel: For any $n \in \mathbb{N}^*$ and $\sigma > 0$, the n -dimensional Gaussian kernel is:

$$K_{\sigma}^{(n)} : (x, y) \in \mathbb{R}^n \times \mathbb{R}^n \mapsto \exp(-\|x - y\|^2 / \sigma) \in \mathbb{R}$$

The proposed kernel: Let K_G be a kernel defined for any $(t, x), (t', x') \in (\mathbb{R}^{d+1})^2$,

$$K_G((t, x), (t', x')) = \begin{pmatrix} c_0 K_{\text{time}} & 0 \\ 0 & c_1 K_{\text{space}} \end{pmatrix} \text{ with } \begin{cases} K_{\text{time}} = K_{\sigma_{T,0}}^{(1)}(t, t') \\ K_{\text{space}} = K_{\sigma_{T,1}}^{(1)}(t, t') K_{\sigma_x}^{(d)}(x, x') I_{\mathbb{R}^n} \end{cases}$$

parametrized by $\sigma_{T,0}, \sigma_{T,1}, \sigma_x > 0$ and the constants $c_0, c_1 > 0$.

Lemma: For any initial velocity field $v_0 \in V$, the RKHS associated to K_G , and for any $\tau \in [0, 1]$, there exist a time parametrization γ_{τ} and a distortion h_{τ} such that

$$\exp_{Id}(\tau v_0) \cdot G(s_0) = G((s_0 + h_{\tau}) \circ \gamma_{\tau}^{-1}).$$

⚠ Time series are not curves:

- ▶ A **curve** $s : t \in [0,1] \mapsto s(t) \in \mathbb{R}^{d+1}$, interest in its **image**: $\{s(t) \mid t \in [0,1]\}$.
- ▶ A **time series** $s : t \in \mathbf{I} \mapsto s(t) \in \mathbb{R}^d$, interest in its **graph**: $\{(t, s(t)) \mid t \in \mathbf{I}\}$.

Shape analysis:

Tailored for curves analysis (independent of time parametrization):

- ▶ LDDMM for curves analysis [Gla08]
- ▶ Square Root Velocity (SRV) [Sri10, Heo24]

Tailored for time series analysis:

- ▶ Functional data analysis (Shape-FPCA) [Wu24]

Deep learning:

- ▶ RNN-Based: [Med99, Chu14]
- ▶ Attention-based: [Shu21, Lee21]
- ▶ Neural-ODEs and its variants: [Lec20, Oh24]

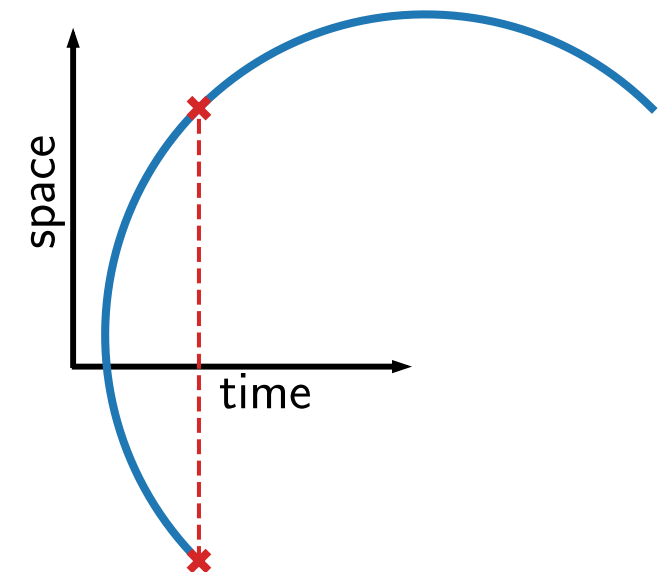


Figure: Difference between curve and time series

Experimental protocol:

- ▶ Task: Classification with different scenarios of irregular sampling.
- ▶ Datasets: 15 UCR/UEA datasets.
- ▶ Metric: f1-score.
- ▶ Algorithms: TS-LDDMM, LDDMM and 8 deep learning algorithms.

Results:

First & second best performers. TS-LDDMM is the best performer on three out of four dropping rate.

Methods	Regular		30 % dropped		50 % dropped		70 % dropped	
	F1-score	Rank	F1-score	Rank	F1-score	Rank	F1-score	Rank
RNN (1999)	0.64 ± 0.21	6.2	0.53 ± 0.23	6.6	0.48 ± 0.21	7.2	0.44 ± 0.21	6.07
LSTM (1997)	0.61 ± 0.29	6.0	0.57 ± 0.29	6.27	0.53 ± 0.25	6.07	0.51 ± 0.29	5.27
GRU (2014)	0.71 ± 0.26	4.2	0.68 ± 0.28	4.27	0.66 ± 0.28	3.73	<u>0.59 ± 0.28</u>	<u>3.67</u>
MTAN (2021)	0.59 ± 0.28	7.13	0.58 ± 0.28	5.8	0.54 ± 0.29	5.33	0.51 ± 0.28	5.0
MIAM (2022)	0.48 ± 0.35	6.93	0.42 ± 0.33	8.27	0.47 ± 0.31	6.93	0.35 ± 0.31	7.6
ODE-LSTM (2020)	0.63 ± 0.24	6.0	0.57 ± 0.25	6.53	0.51 ± 0.24	7.27	0.45 ± 0.23	6.73
Neural SDE (2019)	0.48 ± 0.28	7.67	0.47 ± 0.26	7.47	0.45 ± 0.27	7.13	0.45 ± 0.25	6.0
Neural LNSDE (2024)	0.7 ± 0.27	<u>3.87</u>	0.68 ± 0.29	<u>4.0</u>	<u>0.67 ± 0.25</u>	<u>3.53</u>	0.66 ± 0.23	2.47
LDDMM (2008)	<u>0.72 ± 0.2</u>	4.53	<u>0.7 ± 0.21</u>	4.2	0.57 ± 0.25	5.0	0.4 ± 0.25	7.13
TS-LDDMM (ours)	0.83 ± 0.18	2.93	0.8 ± 0.18	2.07	0.7 ± 0.26	3.33	0.51 ± 0.27	5.67

Mouse ventilation analysis

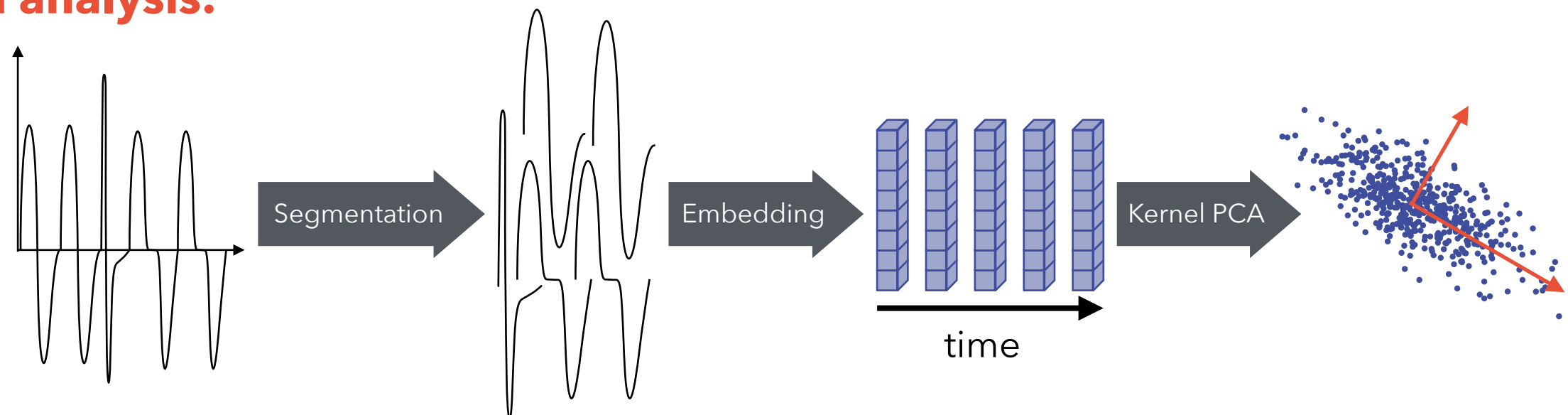


Aim: Assessing ventilation response after the exposure of a neurotoxin for mice of different genotypes (WT and ColQ).

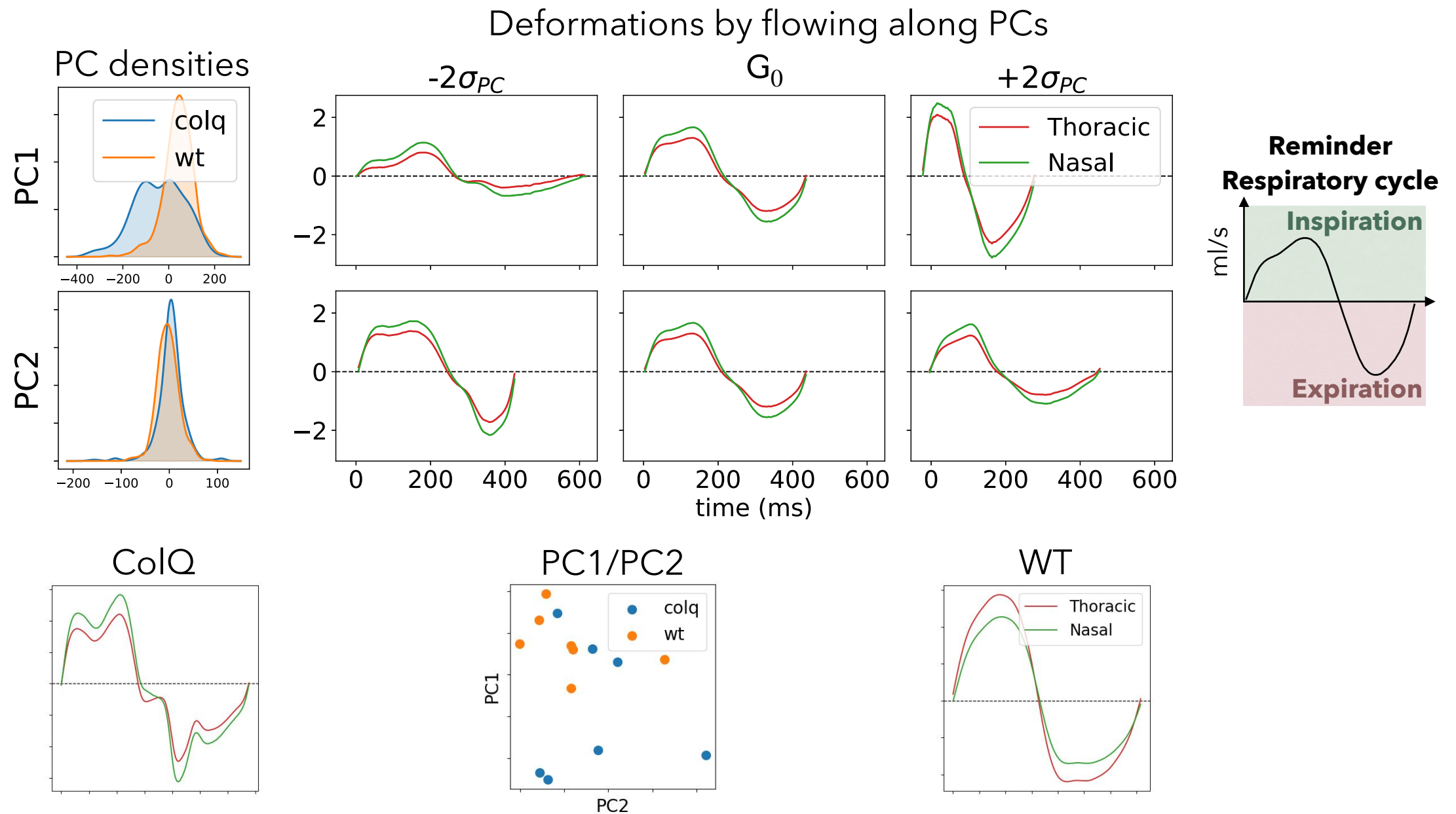
Experimental protocol:



Data analysis:



Kernel PCA applied on the initial velocity fields parametrized with $(G_0, \alpha_j)_{j \in \llbracket 1, N \rrbracket}$.



Observations:

- ▶ PC1 accounts for the cycle duration and PC2 for the ratio inspiration/expiration.
- ▶ PC1 negative tail for ColQ mice seems to highlight impairment of motor control.

Advantages:

- ▶ Interpretable method (Dialogue around the data)
- ▶ Statistical guarantees (representation infectivity)
- ▶ Handles irregularly sampled time series of variable size

Limits:

- ▶ High computational cost.
- ▶ Requires smooth time series.

Future work:

- ▶ Layout the downstream machine learning pipeline.

Thank you for your attention.

Any questions?

► Similarity search:

- Thibaut Germain, Charles Truong, and Laurent Oudre. **"Linear-trend normalization for multivariate subsequence similarity search"**. In: 2024 IEEE 40th International Conference on Data Engineering Workshops (**ICDEW**). IEEE. 2024, pp. 167–175. DOI: [10.1109/ICDEW61823.2024.00028](https://doi.org/10.1109/ICDEW61823.2024.00028)

► Motif discovery:

- Thibaut Germain, Charles Truong, and Laurent Oudre. **"Interactive Motif Discovery in Time Series with Persistent Homology"**. In: Machine Learning and Knowledge Discovery in Databases (**ECML PKDD**). Research Track and Demo Track. Springer. 2024, pp. 383–387. DOI: [10.1007/978-3-031-70371-3_25](https://doi.org/10.1007/978-3-031-70371-3_25)
- Thibaut Germain, Charles Truong, and Laurent Oudre. **"Persistence-Based Motif Discovery in Time Series"**. In: IEEE Transactions on Knowledge and Data Engineering (**TKDE**) 36.11 (2024), pp. 6814–6827. DOI: [10.1109/TKDE.2024.3417303](https://doi.org/10.1109/TKDE.2024.3417303)

► Clustering-based mice ventilation analysis:

- Thibaut Germain, Charles Truong, Laurent Oudre, and Eric Krejci. **"Unsupervised classification of plethysmography signals with advanced visual representations"**. In: Frontiers in Physiology (**FPHYS**) 14 (2023). DOI: [10.3389/fphys.2023.1154328](https://doi.org/10.3389/fphys.2023.1154328)
- Thibaut Germain, Charles Truong, Laurent Oudre, and Eric Krejci. **"Unsupervised study of plethysmography signals through DTW clustering"**. In: 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (**EMBC**). IEEE. 2022, pp. 3396–3400. DOI: [10.1109/EMBC48229.2022.9870907](https://doi.org/10.1109/EMBC48229.2022.9870907)

► Deformation-based time series embedding:

- Thibaut Germain, Samuel Gruffaz, Charles Truong, Alain Durmus, and Laurent Oudre. **"Shape analysis for time series"**. In: The Thirty-eighth Annual Conference on Neural Information Processing Systems (**NeurIPS**). 2024

- [Ala20] Sara Alaei, Kaveh Kamgar, and Eamonn Keogh. 2020. Matrix profile XXII: exact discovery of time series motifs under DTW. In 2020 IEEE international conference on data mining (ICDM). IEEE, 900-905.
- [Bag14] Anthony Bagnall, Jon Hills, and Jason Lines. 2014. Finding motif sets in timeseries. arXiv preprint arXiv:1407.3685 (2014).
- [Bau21] Martin Bauer et al. "Intrinsic riemannian metrics on spaces of curves: Theory and computation". In: Handbook of Mathematical Models and Algorithms in Computer Vision and Imaging: Mathematical Imaging and Vision (2021), pp. 1-35.
- [Beg05] Beg, M. F., Miller, M. I., Trounev, A., & Younes, L. (2005). Computing large deformation metric mappings via geodesic flows of diffeomorphisms. International journal of computer vision, 61, 139-157.
- [Boi24] Bois, A., Tervil, B., & Oudre, L. (2024). Persistence-based clustering with outlier-removing filtration. Frontiers in Applied Mathematics and Statistics, 10, 1260828.
- [Cec22] Matteo Ceccarello and Johann Gamper. 2022. Fast and Scalable Mining of Time Series Motifs with Probabilistic Guarantees. Proceedings of the VLDB Endowment 15, 13 (2022), 3841-3853.
- [Chu14] Junyoung Chung et al. "Empirical evaluation of gated recurrent neural networks on sequence modeling". In: arXiv preprint arXiv:1412.3555 (2014).
- [Deb19] Debavelaere, V., Bône, A., Durrleman, S., Allasonnière, S., & Alzheimer's Disease Neuroimaging Initiative. (2019). Clustering of longitudinal shape data sets using mixture of separate or branching trajectories. In Medical Image Computing and Computer Assisted Intervention-MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13-17, 2019, Proceedings, Part IV 22 (pp. 66-74). Springer International Publishing.
- [Egn16] SE Roian Egnor and Kristin Branson. "Computational analysis of behavior". In: Annual review of neuroscience 39.1 (2016), pp. 217-236.
- [Gau23] Gaudfernau, F. (2023). Analyse quantitative du développement cérébral foetal sain et anormal à l'aide de modèles de déformation (Doctoral dissertation, Université Paris Cité).
- [Gla05] Glaunes, J. (2005). Transport par difféomorphismes de points, de mesures et de courants pour la comparaison de formes et l'anatomie numérique. These de sciences, Université Paris, 13.
- [Gra16] Josif Grabocka, Nicolas Schilling, and Lars Schmidt-Thieme. 2016. Latent time-series motifs. ACM Transactions on Knowledge Discovery from Data (TKDD) 11, 1 (2016), 1-20.
- [Gre94] Grenander, U. (1994). General pattern theory: A mathematical study of regular structures. Oxford University Press.
- [Gre98] Grenander, U., & Miller, M. I. (1998). Computational anatomy: An emerging discipline. Quarterly of applied mathematics, 56(4), 617-694.
- [Gru21] Gruffaz, S., Poulet, P. E., Maheux, E., Jedynak, B., & Durrleman, S. (2021). Learning Riemannian metric for disease progression modeling. Advances in Neural Information Processing Systems, 34, 23780-23792.

- [Jos16] Josif Grabocka, Nicolas Schilling, and Lars Schmidt-Thieme. 2016. Latent time-series motifs. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 11, 1(2016), 1-20.
- [Kal17] Kaltenmark, I., Charlier, B., & Charon, N. (2017). A general framework for curve and surface comparison and registration with oriented varifolds.
- [Ken77] Kendall, D. G. (1977). The diffusion of shape. *Advances in applied probability*, 9(3), 428-430.
- [Kli15] Christian Peter Klingenberg. "Analyzing fluctuating asymmetry with geometric morphometrics: concepts, methods, and applications". In: *Symmetry* 7.2 (2015), pp. 843-934
- [Lec20] Mathias Lechner and Ramin Hasani. "Learning long-term dependencies in irregularly-sampled time series". In: *arXiv preprint arXiv:2006.04418* (2020).
- [Lee21] Yurim Lee et al. "Multi-view integrative attention-based deep representation learning for irregular clinical time-series data". In: *IEEE Journal of Biomedical and Health Informatics* 26.8 (2022), pp. 4270-4280.
- [Li15] Yuhong Li, Hou U Leong, Man Lung Yiu, and Zhiguo Gong. 2015. Quick-motif: An efficient and scalable framework for exact motif discovery. In *2015 IEEE 31st International Conference on Data Engineering*. IEEE, 579-590.
- [Lin18] Michele Linardi, Yan Zhu, Themis Palpanas, and Eamonn Keogh. 2018. Matrixprofile X: VALMOD-scalable discovery of variable-length motifs in data series. In *Proceedings of the 2018 international conference on management of data*. 1053-1066
- [Mar17] Lucie Maršánová et al. "ECG features and methods for automatic classification of ventricular premature and ischemic heartbeats: A comprehensive experimental study". In: *Scientific reports* 7.1 (2017), p. 11239.
- [Med99] Larry Medsker and Lakhmi C Jain. *Recurrent neural networks: design and applications*. CRC press, 1999.
- [Mil06] Miller, M. I., Trounev, A., & Younes, L. (2006). Geodesic shooting for computational anatomy. *Journal of mathematical imaging and vision*, 24, 209-228.
- [Mue09] Abdullah Mueen, Eamonn Keogh, Qiang Zhu, Sydney Cash, and Brandon Westover. 2009. Exact discovery of time series motifs. In *Proceedings of the 2009 SIAM international conference on data mining*. SIAM, 473-484.
- [Oh24] YongKyung Oh, Dongyoung Lim, and Sungil Kim. "Stable Neural Stochastic Differential Equations in Analyzing Irregular Time Series Data". In: *The Twelfth International Conference on Learning Representations*. 2024.
- [Rak12] Thanawin Rakthanmanon, Eamonn J Keogh, Stefano Lonardi, and Scott Evans. 2012. MDL-based time series clustering. *Knowledge and information systems* 33(2012), 371-399.

- [Sar20] Sara Alaei, Kaveh Kamgar, and Eamonn Keogh. 2020. Matrix profile XXII: exact discovery of time series motifs under DTW. In 2020 IEEE International Conference on Data Mining (ICDM). IEEE, 900–905.
- [Sen18] Pavel Senin, Jessica Lin, Xing Wang, Tim Oates, Sunil Gandhi, Arnold P Boedi-hardjo, Crystal Chen, and Susan Frankenstein. 2018. Grammarviz 3.0: Interactive discovery of variable-length time series patterns. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 12, 1 (2018), 1–28.
- [Shu21] Satya Narayan Shukla and Benjamin M Marlin. “Multi-time attention networks for irregularly sampled time series”. In: *arXiv preprint arXiv:2101.10318* (2021).
- [Tho17] J Arthur Thomson. *On growth and form*. 1917.
- [Wu24] Yuexuan Wu, Chao Huang, and Anuj Srivastava. “Shape-based functional data analysis”. In: *TEST* 33.1 (2024), pp. 1–47.
- [Yeh16] Chin-Chia Michael Yeh, Yan Zhu, Liudmila Ulanova, Nurjahan Begum, Yifei Ding, Hoang Anh Dau, Diego Furtado Silva, Abdullah Mueen, and Eamonn Keogh. 2016. Matrix profile I: all pairs similarity joins for time series: a unifying view that includes motifs, discords and shapelets. In 2016 IEEE 16th International Conference on Data Mining (ICDM). IEEE, 1317–1322.
- [You10] Younes, L. (2010). *Shapes and diffeomorphisms* (Vol. 171, pp. xviii+434). Berlin: Springer.
- [Yuh15] Yuhong Li, Hou U Leong, Man Lung Yiu, and Zhiguo Gong. 2015. Quick-motif: An efficient and scalable framework for exact motif discovery. In 2015 IEEE 31st International Conference on Data Engineering. IEEE, 579–590.